MAWLANA BHASHANI SCIENCE AND TECHNOLOGY UNIVERSITY

DEPARTMENT OF INFORMATION AND COMMUNICATION TECHNOLOGY

PRESENTATION ON

RIP: Routing Information Protocol

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Course: **Data Communication**

Course Code: **ICT-3101**

Topic: RIP

Presentation roadmap

01

01 Foundations

Why routers need a dynamic routing protocol

02

02 RIP in action

Distance vectors, hop counts and route updates

03

03 Versions and resilience

RIPv1, RIPv2, RIPng and loop prevention

04

04 Evaluation

Advantages, limitations and protocol comparison

The routing problem

Many networks

Routers must discover paths to destinations beyond directly connected links.

Networks change

Links can fail, new networks appear and manual updates become difficult.

A shared solution

Dynamic routing lets routers exchange reachability information.

What is RIP?

01

Distance-vector routing protocol

Routers advertise destination networks and their distance in hops.

02

Bellman–Ford principle

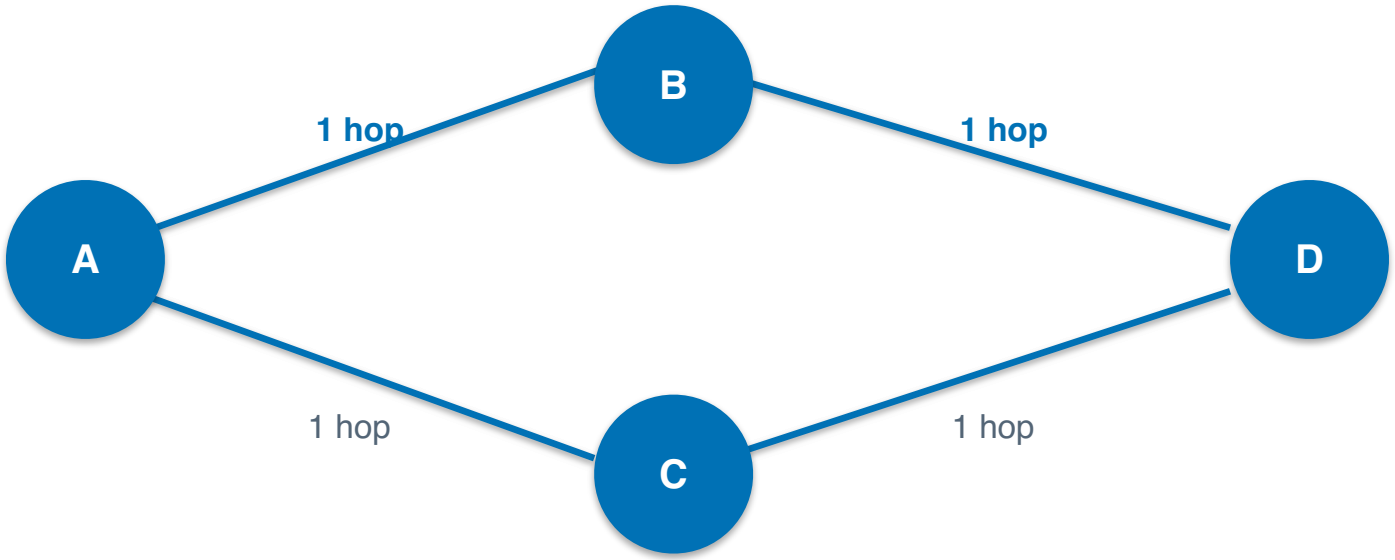
Each router estimates the best route using information from neighbours.

03

Interior gateway protocol

Designed for routing within a single administrative domain.

RIP topology I Finding the shortest path



A → D

via B: 2 hops

via C: 2 hops

Illustrative topology: RIP compares hop count, not bandwidth or delay.

How RIP learns a route

01

Step 1 | Discover

Each router knows its directly connected networks.

02

Step 2 | Advertise

Neighbours exchange routing entries, typically every 30 seconds.

03

Step 3 | Update

A router adds one hop and selects a route with the lowest hop count.

04

Step 4 | Adapt

Changed or unreachable routes trigger routing-table updates.

A simple routing-table example

Destination	Next hop	Hop count
192.168.10.0/24	Direct	0
192.168.20.0/24	Router B	1
192.168.30.0/24	Router B	2
192.168.40.0/24	Router C	3

Illustrative entries. Connected networks are shown as 0 hops locally.

RIP essentials

Metric

Hop count only; the route with the fewest hops is preferred.

Maximum reach

15 hops is the maximum usable metric; 16 means unreachable.

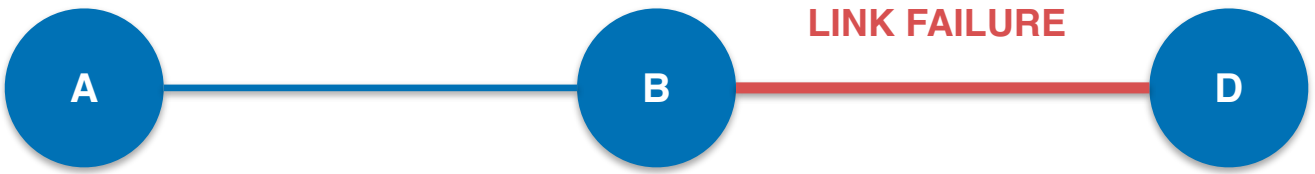
Updates

Periodic routing advertisements, normally every 30 seconds.

Transport

RIPv1 and RIPv2 use UDP port 520.

When a link fails | Convergence



1 Route becomes unavailable

2 Router advertises the change

3 Neighbours recalculate their tables

RIP convergence can be slow, especially in larger or unstable networks.

How RIP reduces routing loops

Split horizon

Do not advertise a route back through the interface from which it was learned.

Route poisoning

Advertise a failed destination with metric 16 (unreachable).

Triggered updates

Send an update promptly after a significant routing change.

RIP versions | At a glance

Feature	RIPv1	RIPv2	RIPng
Address family	IPv4	IPv4	IPv6
Classless / CIDR	No	Yes	Yes
Update delivery	Broadcast	Multicast	Multicast
Authentication	No	Supported	Uses IPv6 security mechanisms
UDP port	520	520	521

RIPng is the IPv6 adaptation of RIP; it is not simply RIPv2 with longer addresses.

Advantages and limitations

01

Why RIP is still useful

Easy to understand, simple to configure and appropriate for small networks or labs.

02

Where RIP struggles

15-hop limit, slow convergence and no bandwidth-aware path selection.

03

Practical takeaway

A straightforward teaching protocol, but often not the first choice for large modern networks.

RIP vs OSPF vs EIGRP

Aspect	RIP	OSPF	EIGRP
Approach	Distance vector	Link state	Advanced distance vector
Main metric	Hop count	Cost	Composite metric
Scalability	Small networks	Large networks	Medium to large
Convergence	Relatively slow	Generally fast	Generally fast
Complexity	Low	Higher	Moderate

Performance depends on topology, implementation and configuration.

Where RIP makes sense

Networking labs

Learn routing tables, advertisements, metrics and convergence.

Small legacy networks

Simple environments with limited size and modest routing requirements.

Protocol fundamentals

A clear introduction before studying OSPF, IS-IS or BGP.

Key takeaways

01

RIP exchanges distance vectors

Routers learn routes through advertisements from their neighbors.

02

Hop count drives decisions

15 hops is the maximum usable distance; 16 means unreachable.

03

Simplicity has trade-offs

RIP is accessible but less scalable and slower to converge than modern alternatives.

Selected references

[1] RFC 1058 — Routing Information Protocol (RIPv1), C. Hedrick, 1988.

[2] RFC 2453 — RIP Version 2, G. Malkin, 1998.

[3] RFC 2080 — RIPng for IPv6, G. Malkin and R. Minnear, 1997.

[4] RFC 4822 — RIPv2 Cryptographic Authentication, R. Atkinson and M. Fanto, 2007.

THANK YOU